

Investing in Quantum: A High-Risk, High-Reward Bet on the Future

Part 1 of 5: Thoughts on Quantum Technology Based on Interpretations at SEEQA Quantum Conference Hosted at Oxford University



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Image captured at Oxford University during the SEEQA 2025 Conference

I have just come back from SEEQA – a quantum conference hosted at Oxford University. I had an absolutely lovely time meeting and learning from prominent quantum researchers. We even got to watch an Oxford Union debate—an absolute treat. I had many interesting conversations, but the ones that stood out most for me were centred on the funding flowing into quantum and the industrial economics of it all.

Below is research informed by those conversations, along with my take on where quantum is heading.

Where the Money Is Flowing

The industry remains largely in a pre-revenue phase, but funding momentum has rebounded sharply. After cooling in 2022–23, quantum startups raised nearly \$2 billion in 2024, up ~50% from the prior year. Strikingly, governments now account for more than one-third of startup funding, reflecting the technology's strategic importance.

Venture dollars, meanwhile, have been highly selective:

- 48% of investment in 2023 went to hardware—superconducting, photonic, silicon, and trapped-ion systems that demand cryogenics and high-end labs.
- Only ~6% went to software, underscoring that progress still depends on getting physical qubits right.
- Among hardware bets, silicon qubits drew ~38% of qubit-focused investment, photonic 27%, and neutral atoms 20%.

In other words, venture capital is betting on the physics that will scale, not just on who builds the next app. Hardware has become the gravitational center of quantum investment, with software and sensing viewed as “low-risk” ways to monetize in the interim.

The Industrial Economics of Quantum

Quantum's economics look nothing like SaaS or fintech. They resemble the **early semiconductor industry**—heavy infrastructure, long payback periods, and reliance on state funding.

Capital Intensity

Building quantum systems is brutally expensive. A single dilution refrigerator for superconducting qubits costs upwards of \$1 million, and that's before lasers, vacuum chambers, cryogenic cabling, and cleanroom-grade fabrication. Even running prototypes requires electricity-intensive cooling plants and specialized technicians. Compare this to a cloud startup, where costs are server time and salaries; in quantum, infrastructure is closer to aerospace or biotech.

Concentration of Capital

Nearly half of recent global funding went to hardware, with most venture dollars chasing a handful of modalities. Silicon qubits (championed by Australia's Silicon Quantum Computing and Diraq) dominate, followed by photonic players like Xanadu and PsiQuantum, and neutral-atom ventures such as Pasqal. Investors are making **physics bets**, not product bets, on which qubit type will survive the coming shakeout.

Barriers to Entry

Unlike consumer tech, quantum has structural barriers that discourage copycats. Startups need teams of PhDs in quantum physics, years of specialized training, and access to scarce materials. Even software-focused firms must secure privileged access to hardware through partnerships with IBM, AWS, or IonQ. Without such access, algorithmic innovations can't be tested at scale.

Revenue Models and Runway

Most quantum companies have little to sell today beyond cloud access, consulting, and government contracts. Enterprise-grade products—dr

simulation platforms, secure quantum networks, optimization engines remain **five to ten years away**. As a result, companies report modest revenue (tens of millions) against massive R&D burn. This looks more like biotech or traditional enterprise software: years of research and negative cash flow followed by binary moments of validation when breakthroughs arrive.

Public Market Volatility

Despite thin revenues, several quantum firms are already public. IonQ has a **\$12 billion valuation** after a fivefold stock run-up, while D-Wave's market cap climbed above \$5 billion. Yet these valuations are volatile, swinging hundreds of percent in a short window. For investors, this is less about current cash flows and more about optionality—the possibility that a quantum bet pays off once error correction and scaling are solved.

Role of the State

No other deep-tech sector is as interwoven with national strategy. The US National Quantum Initiative, Europe's €1 billion Quantum Flagship, and China's \$10 billion+ state push ensure governments are **both financiers and customers**. Defence contracts, supercomputing centers, and telecom networks are the early adopters. For startups, survival often depends on securing government grants or partnerships that subsidize years of unprofitable R&D.

Takeaway: This is a patient-capital game. Industrial economics reward those who combine **deep-pocketed partners** (corporates, governments) with **niche revenues** (sensing, cryptography) to extend their runway until large scale systems emerge.

Use Cases on the Horizon

The industrial economics matter because they shape how use cases roll out. Near-term revenue comes from applications that align with defence, pharma, and financial sectors willing to co-fund R&D. Longer-term returns hinge on scaling error-corrected machines.

Pharmaceuticals and Chemistry

Quantum promises breakthroughs in drug discovery and materials science. By simulating complex molecules with quantum algorithms, companies like Quantinuum (U.S.) and Quantinuum (Germany) and Zapata Computing (U.S.) are already working with pharma giants. Even today's noisy devices can model small molecules as proofs of concept, while future systems could accelerate pipelines.

Defence and National Security

Governments see both opportunity and threat. Quantum sensors could detect stealth submarines or aircraft, while quantum inertial navigation provides GPS-free guidance. Quantum cryptography (QKD) is already deployed in telecoms, securing networks in China, Switzerland, and South Korea. These dual-use technologies explain why public funding is so central: they are matters of national security as much as innovation.

Finance and Logistics

Quantum-inspired algorithms are already being tested in portfolio optimization, risk modelling, and supply chain planning. Startups like Multiverse Computing (Spain) are running pilot projects with banks and

manufacturers, while logistics firms are experimenting with annealers (D-Wave) for routing and scheduling. These applications don't require full error-corrected systems, making them attractive for near-term commercialization.

Telecommunications

The race to build the quantum internet is real. Secure QKD links are live in Switzerland and South Korea, while satellite-based QKD is being tested by Space Quantum (Singapore). These infrastructures could one day underpin global communications the way undersea cables did in the 19th century.

Healthcare Imaging

Quantum sensors—like NV-diamond magnetometers—could one day deliver molecular-level MRI scans, transforming diagnostics and personalized medicine. Early prototypes exist, but the industrial economics mean that these will require partnerships between hospitals, universities, and quantum labs.

The Bigger Picture

The addressable market is still nascent—roughly \$650–750 million in revenue in 2024, projected to exceed \$1 billion in 2025. Yet forecasts suggest \$9 billion in revenue potential by 2035 across computing, communications, and

Investing in quantum today is not about quarterly earnings. It's about underwriting the physics of the future. The winners will be those who manage extraordinary burn rates, navigate geopolitical jockeying, and early industrial use cases.

In other words, quantum is a moonshot market. High barriers, high risk if the hype holds—returns that could reshape entire industries.

Question for readers: If you had to allocate capital, would you treat quantum like semiconductors in the 1950s—an infrastructure bet with generational upside—or like biotech in the 1990s, where many firms failed but a few rewrote the future?

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